

Exceptional-Point Stability Boundaries from Quantum Dissipation to Cosmological Acceleration

Nate Christensen

SymC Universe Project, Missouri, USA

NateChristensen@SymCUniverse.com

09 February 2026 (revised April 2026)

Abstract

Dissipative dynamics across physical systems exhibit organizing structural boundaries. The dimensionless damping ratio $\chi \equiv \gamma/(2\omega)$ defines a stability architecture in which the critical threshold $\chi = 1$ marks a second-order non-Hermitian Exceptional Point (EP2). Within spatially flat Λ CDM, an exact algebraic identity is demonstrated linking the onset of cosmic acceleration to critical damping of structure growth: $\chi_\delta = 1 \iff q = 0$. This identity is a structural reformulation of the standard Friedmann and growth equations, not a new physical law; its scientific value lies in recasting a known kinematic transition as a stability-phase transition and in generating the falsifiable prediction that departures from flat Λ CDM produce a measurable, nonzero offset between the $q = 0$ and $\chi_\delta = 1$ transitions (identically zero under flat Λ CDM, generically nonzero in extensions). A substrate inheritance relation is then proposed as a leading-order projection approximation, whereby emergent modes acquire effective parameters from substrate precursors. Within this framework, the observed particle distribution is consistent with stability organization: long-lived matter occupies localized stability basins at $\chi \ll 1$, while short-lived resonances occupy a secondary $\chi \sim 0.1$ – 1 band whose sub-cluster evidence is suggestive rather than conclusive. The framework is presented as a structurally grounded classification with defined scope and testable consequences.

1 Boundary Structure in Dissipative Dynamics

The goal of this work is not to propose new microphysical laws but to identify structural commonalities in the mathematical form of dissipative dynamics across domains. The organizing object is a single dimensionless ratio whose qualitative role is identical in every system admitting an effective second-order response.

Many physical theories are structured around boundary conditions and limiting principles. The speed of light c limits information propagation, and Planck's constant $\hbar$ bounds measurement precision. The dimensionless ratio

$$\chi \equiv \frac{\gamma}{2\omega} \tag{1}$$

plays an analogous role in dynamical stability. Here ω denotes the positive characteristic frequency scale of the mode. Among dimensionless combinations of γ and ω , χ uniquely controls the sign of the quadratic discriminant $\Delta = \gamma^2 - 4\omega^2$ and therefore the qualitative class of the system's

eigenvalues. It is the minimal invariant parameter governing stability classification: $\Delta > 0$ iff $\chi > 1$ (overdamped); $\Delta = 0$ iff $\chi = 1$ (critically damped); $\Delta < 0$ iff $\chi < 1$ (underdamped). Any other dimensionless combination of γ and ω can be written as a function of χ alone. Across all domains discussed in this manuscript, χ is defined as the ratio of effective damping to twice the characteristic response frequency; domain-specific mappings (e.g., $\Gamma \leftrightarrow \gamma$, $m \leftrightarrow \omega$ for particle physics) are interpretations of this common definition rather than distinct definitions.

At $\chi = 1$, the quadratic response has a repeated root; in defective non-Hermitian realizations the effective generator coalesces and the impulse response transitions to

$$h(t) = te^{-\omega t}, \quad (2)$$

defining a second-order non-Hermitian exceptional point (EP2) [1, 11–13]. The term “exceptional point” is used throughout in the restricted algebraic sense of discriminant coalescence in the quadratic response kernel; specific non-Hermitian realizations require that the relevant reduced generator possess the corresponding defective structure.

Quadratic eigenproblems for linear response arise across classical dissipative modes, retarded propagators in open quantum field theory, and cosmological growth dynamics [4, 6, 9, 17]. Markovian open-system generators of Lindblad type can realize the same EP2 structure in suitable reduced response channels or effective mode equations (Supplementary Section 1) [10, 14], but the EP condition is asserted here at the level of the shared quadratic response structure.

Physical interpretation and scope

Definitions. The term *stability architecture* denotes the partitioning of effective-mode parameter space by the critical-damping boundary $\chi = 1$ into underdamped ($\chi < 1$, oscillatory decay), critically damped ($\chi = 1$, algebraic-exponential kernel Eq. (2)), and overdamped ($\chi > 1$, monotone exponential decay) sectors. The term *structural separatrix* denotes the codimension-one surface $\chi = 1$ in the (ω, γ) plane at which eigenvalues coalesce and response changes mathematical class from oscillatory to monotone decay.

Cross-domain scope. The correspondences drawn across domains are structural, arising from shared mathematical form in the effective second-order response, rather than analogical in the sense of assuming physical equivalence between systems. The framework applies only to systems admitting a dominant-mode reduction to a second-order effective equation; strongly multimodal, chaotic, or intrinsically non-Markovian systems do not in general admit such a reduction without additional coarse-graining.

Utility. The utility of the framework lies in converting qualitative regime distinctions into a single measurable parameter, enabling cross-system comparison and identification of structural transitions not explicit in the original formulations of each domain. The Δz offset in cosmology, the spectral doublet merger in circuit QED, and the viscous-dissipation shift in neutron stars are all consequences of this single-parameter classification.

Epistemic classification. The results span three epistemic levels kept distinct throughout. (1) *Derived results:* the cosmological identity $\chi_\delta = 1 \iff q = 0$ (exact within flat Λ CDM), the EP2 discriminant structure (exact for the specified second-order response class), and the perturbative RG stability estimate for weakly coupled $\lambda\phi^4$ theory. (2) *Structural analogies supported by formal consistency:* the substrate inheritance relation (a structural hypothesis motivated by consistency across reduced descriptions, not derived from QFT), the particle stability classification, and the

Table 1: Epistemic classification of claims.

Level	Claim	Basis
Derived (algebraic)	$\chi_\delta = 1 \Leftrightarrow q = 0$	Friedmann and growth equations.
Derived (control theory)	EP2 at $\chi = 1$	Quadratic eigenvalue theory.
Retracted	Efficiency maximum near $\chi = 1$ for tested proxies	Full numerical integration; Lemma S2.1 conditional only.
Empirical (observational)	Particle χ values from PDG masses and widths	PDG 2022 measurements.
Constructed (by definition)	Substrate $\chi = 1$ assignments	Imposed by construction; excluded from statistical tests.
Structural hypothesis	Substrate inheritance relation	Consistency argument in reduced descriptions; not a QFT derivation.
Conjectural	Gauge couplings as substrate ratios	Speculative extension.
Retracted	NS $\chi_{\text{NS}} \rightarrow 1$ near M_{max}	Inconsistent with published QNM data.

information-efficiency interpretation for specified channel models. (3) *Speculative extensions requiring future calculation*: the QCD substrate χ estimate, the gauge-coupling stability classification, and quantitative cross-domain structural claims. Table 1 summarizes the epistemic status of each major claim.

2 Exceptional-Point Kernel and Efficiency Scope

For a representative linear mode whose observable response obeys a second-order dissipative equation:

$$\ddot{x} + \gamma\dot{x} + \omega^2x = 0, \quad (3)$$

the characteristic discriminant

$$\Delta = \gamma^2 - 4\omega^2 = 4\omega^2(\chi^2 - 1) \quad (4)$$

vanishes at $\chi = 1$, yielding coalesced roots $\lambda_\pm = -\omega$ and the EP2 impulse kernel (2) [12]. Such quadratic response structures appear directly in classical linear-response theory and in retarded propagators of dissipative quantum field theory [4]. Lindblad-form Markovian dynamics can realize the same EP2 kernel in appropriate effective response channels; the mapping and its scope are given explicitly in Supplementary Section 1 [10, 14].

An identical structure arises in dissipative quantum field theory. For a scalar mode with retarded propagator

$$G_R(\Omega) = \frac{1}{-\Omega^2 - i\gamma\Omega + \omega^2}, \quad (5)$$

the poles

$$\Omega_\pm = -\frac{i\gamma}{2} \pm \sqrt{\omega^2 - \frac{\gamma^2}{4}} \quad (6)$$

coalesce at $\gamma = 2\omega$, reproducing the same EP2 kernel [4, 13].

The information-efficiency functional

$$\eta(\chi) \equiv \frac{I(\chi)}{\Sigma(\chi)} \quad (7)$$

is model-dependent. Full numerical integration of $\eta(\chi)$ for two representative entropy-production proxy definitions (output-power: $\Sigma \propto \gamma \int |H|^2 d\Omega$; and kinetic: $\Sigma \propto \gamma \int \Omega^2 |H|^2 d\Omega$) shows no interior maximum near $\chi = 1$ for any tested bandwidth ratio $B/\omega_0 \in [0.5, 10]$. In the tested configurations, the output-power proxy and the kinetic proxy at most bandwidths peak at the lower scan boundary (strongly underdamped); the kinetic proxy at $B/\omega_0 = 1$ increases monotonically through the full scan range with no interior turning point. In neither case does an extremum occur near $\chi = 1$. The claim $\chi^* \approx 1.02$ from the original approximate formula is therefore retracted for these proxy definitions (Supplementary Section 2.4). Lemma S2.1 in Supplementary Section 2 remains valid as a conditional localization theorem: if a local maximum of an efficiency functional *exists* at $\chi = 1$ for some channel specification, small perturbations (finite memory, finite bandwidth, measurement filtering) keep it within an $O(\varepsilon/\kappa)$ neighborhood of $\chi = 1$. The strongest statement available is the discriminant boundary itself: systems admitting the same second-order response form share the same stability transition at $\chi = 1$ regardless of any channel model or proxy choice.

Physically, $\chi < 1$ corresponds to persistent ringing, while $\chi > 1$ sacrifices responsiveness [15]. Figure 1 visualizes this dual structure as an exceptional-point manifold separating overdamped and underdamped regimes.

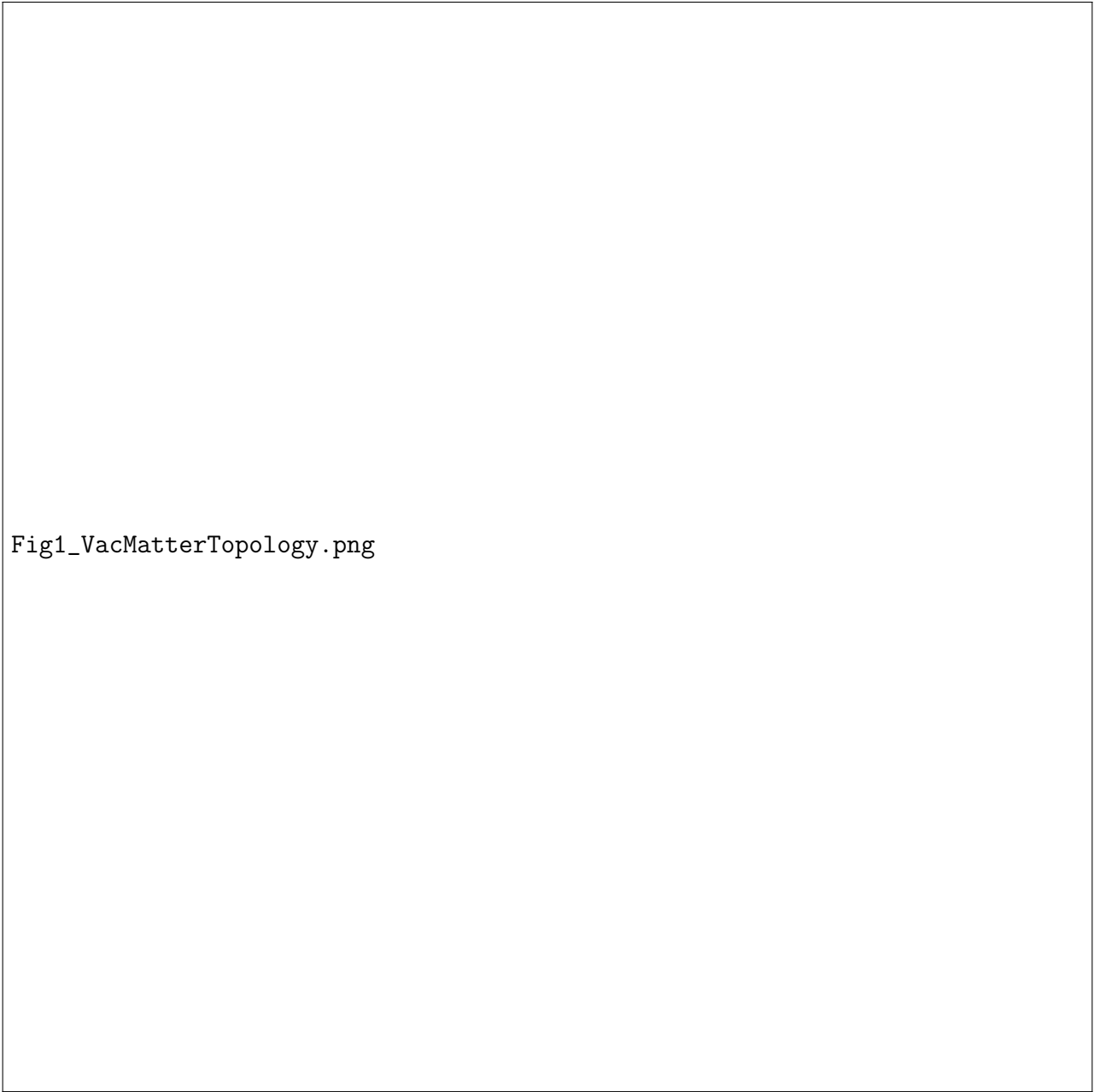


Fig1_VacMatterTopology.png

Figure 1: **Stability manifold and dynamical regime classification.** Three-dimensional visualization in $(\log_{10} \omega, \log_{10} \Gamma, \chi)$ space where $\chi = \Gamma/(2\omega)$ is the dimensionless damping ratio. The colored surface represents the exceptional point (EP) manifold at $\chi = 1$, separating overdamped ($\chi > 1$, magenta) and underdamped ($\chi < 1$, blue) dynamical regimes. Representative systems illustrate an ordering by observed stability index: vacuum-substrate reference anchors are defined at the critical boundary ($\chi = 1$ by construction; see Figure 3), while long-lived excitations lie deep in the underdamped basin. The schematic language is classificatory and does not assert a microscopic pathway. Vertical projection lines connect the 3D schematic to the empirical classification plane (Figure 3).

Structural interpretation

The exceptional-point boundary at $\chi = 1$ is not interpreted as a dynamical attractor. Rather, it serves as a structural separatrix where impulse response transitions from oscillatory to monotone

decay. The framework classifies near-critical operation as a structurally significant operating point, with departures incurring complementary penalties: $\chi < 1$ produces persistent ringing that degrades signal fidelity under finite bandwidth, while $\chi > 1$ produces slow relaxation that suppresses information throughput for fixed dissipation.

3 Cosmic Crossing: $\chi_\delta = 1 \iff q = 0$

3.1 Derived identity within flat Λ CDM

Linear density perturbations satisfy

$$\ddot{\delta} + 2H\dot{\delta} - 4\pi G\rho_m\delta = 0 \quad (8)$$

[9, 17], yielding

$$\chi_\delta = \frac{H}{\sqrt{4\pi G\rho_m}}. \quad (9)$$

Substituting $4\pi G\rho_m = \frac{3}{2}H^2\Omega_m$ and cancelling H exactly yields

$$\chi_\delta(z) = \sqrt{\frac{2}{3\Omega_m(z)}}, \quad (10)$$

a pure function of the evolving matter density parameter. In flat Λ CDM, the deceleration parameter

$$q = \frac{1}{2}\Omega_m - \Omega_\Lambda \quad (11)$$

vanishes at $\Omega_m = 2/3$. The Friedmann equations then give $H^2 = 4\pi G\rho_m$, implying the exact identity

$$\chi_\delta = 1 \iff q = 0. \quad (12)$$

The full derivation is in Supplementary Section 1.3.

3.2 Interpretation and scope

This identity is a reformulation of standard flat- Λ CDM relations. It does not introduce new dark-energy microphysics and is not a new physical law. The coefficients $\gamma_\delta = 2H$ (Hubble friction on peculiar velocities) and $\omega_\delta^2 = \frac{3}{2}H^2\Omega_m$ (gravitational restoring term) are fixed by general relativity rather than free parameters. The structural observation is that general relativity constrains these coefficients to satisfy $\chi_\delta = 1$ exactly when the universe transitions from decelerated to accelerated expansion.

Equation (12) holds exactly for spatially flat Λ CDM. In non-flat cosmologies or in w CDM with $w \neq -1$, the synchronization is no longer exact and the stability crossing $\chi_\delta = 1$ shifts to a nearby redshift. This makes departures testable as measurable offsets between the $q = 0$ and $\chi_\delta = 1$ transitions: the offset

$$\Delta z = z_{q=0} - z_{\chi_\delta=1} \quad (13)$$

is identically zero in flat Λ CDM but generically nonzero in extensions, with $|\Delta z| \approx 0.10\text{--}0.25$ for $w \in \{-0.9, -0.8\}$ and $w = -1.2$ at fixed $\Omega_{m0} = 0.315$. Panel (b) of Figure 2 visualizes this shift in (χ_δ, q) space for representative $w \neq -1$ trajectories.

In summary: $\Delta z = 0$ in flat Λ CDM, and any statistically significant $\Delta z \neq 0$ is a direct indicator of departure from the Λ CDM framework.

3.3 Observational constraints

For concordance parameters $\Omega_{m0} = 0.315$ and $\Omega_{\Lambda0} = 0.685$, the crossing occurs at $z_t = (2\Omega_{\Lambda0}/\Omega_{m0})^{1/3} - 1 \approx 0.632$. Propagating the Planck 2018 uncertainty ($\sigma_{\Omega_m} = 0.007$) gives

$$z_t = 0.632 \pm 0.018 \quad (1\sigma, \text{ Planck 2018}), \quad (14)$$

with 95% CI $[0.598, 0.668]$ from Monte Carlo propagation over 2×10^5 samples. DESI 2024 BAO alone yields $\Omega_{m0} = 0.295 \pm 0.015$ [8], consistent with $z_t = 0.684 \pm 0.040$. Independent kinematic determination from Type Ia supernovae gives $z_t = 0.60^{+0.06}_{-0.08}$ [25], consistent across all datasets within $1-2\sigma$.

Fig2_Synchronization.png

Figure 2: **Cosmological stability crossing and its departures under extended dark energy.** **(a)** Temporal evolution showing the deceleration parameter $q(z)$ (blue) and stability index $\chi_\delta(z)$ (magenta dashed) as functions of redshift, computed directly from the flat- Λ CDM Friedmann and growth equations with concordance parameters $\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$ [19]; no fitting functions are used. Both curves cross at $z_t = 0.632 \pm 0.018$ (1σ , Planck 2018), where $q = 0$ and $\chi_\delta = 1$ simultaneously; this crossing is an algebraic consequence of the flat- Λ CDM equations and serves as the $\Delta z = 0$ reference point for testing extensions. **(b)** Observable Δz offset for w CDM extensions, shown as a trajectory in (χ_δ, q) space. The star marks the flat Λ CDM reference point ($\chi_\delta = 1, q = 0$), at which $\Delta z = 0$ exactly. The solid curve shows the Λ CDM trajectory ($w = -1$); dashed curves show representative $w \neq -1$ models ($w = -0.8, w = -1.2$), which displace the crossing away from the reference point and produce a nonzero Δz . Any statistically significant measured $\Delta z \neq 0$ is a direct indicator of departure from flat Λ CDM.

Within flat Λ CDM, this result involves no free parameters and is independent of any cosmological origin model [18–20].

4 Robustness Under Interactions and Memory

In weakly interacting $\lambda\phi^4$ theory, renormalization-group flow yields

$$\frac{d\chi}{d\ell} = (a_\gamma - a_\omega)\lambda\chi, \quad (15)$$

so that χ remains slowly varying when $a_\gamma \approx a_\omega$ [22]. One-loop calculations confirm $|\Delta\chi| < 0.3\%$ over three decades, with two-loop corrections of order 3×10^{-5} (Supplementary Section 6) [23].

These perturbative results confirm slow running only within weakly coupled models. The one-loop Yukawa correction $d\chi/d(\ln\mu) \sim y^2/(16\pi^2)$ gives $\approx 5 \times 10^{-14}$ per e -fold for the electron ($y_e \approx 2.9 \times 10^{-6}$, entirely negligible) and $\approx 6 \times 10^{-3}$ per e -fold for the top quark ($y_t \approx 1$, non-negligible). Inheritance stability cannot therefore be assumed to hold across all sectors of the Standard Model. Extension to strongly coupled or non-perturbative dynamics, including the QCD critical region, requires dedicated lattice or functional RG analysis; this constitutes an explicit limitation of the present formalism.

Finite-memory baths with kernel $K(t) = \gamma_0 e^{-t/\tau}$ yield an effective damping

$$\gamma_{\text{eff}}(\omega) = \frac{\gamma_0}{1 + (\omega\tau)^2}, \quad (16)$$

broadening but not destroying the critical boundary [4, 7] (Supplementary Section 10 for fractional and power-law extensions). The physically realized damping ratio in such environments is therefore

$$\chi_{\text{phys}}(\omega) \equiv \frac{\gamma_{\text{eff}}(\omega)}{2\omega} = Z(\omega\tau) \chi_{\text{bare}}, \quad Z(\omega\tau) = \frac{1}{1 + (\omega\tau)^2} < 1, \quad (17)$$

so that the exceptional point at $\chi_{\text{bare}} = 1$ corresponds to a downward-shifted operating point when expressed in terms of effective rates.

5 Substrate Inheritance as a Coarse-Grained Structural Relation

5.1 Leading-order projection relation

Let $\{\phi_k\}$ denote substrate modes with frequencies Ω_k and damping Γ_k . Under a leading-order linear projection approximation, any proto-mode $|\psi\rangle = \sum_k c_k |\phi_k\rangle$ inherits effective parameters

$$\Omega_\psi^2 = \sum_k |c_k|^2 \Omega_k^2, \quad \Gamma_\psi = \sum_k |c_k|^2 \Gamma_k. \quad (18)$$

The resulting damping ratio is

$$\chi_\psi = \frac{\Gamma_\psi}{2\sqrt{\Omega_\psi^2}}. \quad (19)$$

This relation is presented as a structural hypothesis motivated by consistency across reduced descriptions; it is not derived from quantum field theory or renormalization-group formalism, and is exact only for the stated linear, diagonal, weak-mixing approximation. In realistic interacting field

theories, off-diagonal mixing, running couplings, thresholds, and non-perturbative effects modify the effective parameters.

To make the overlap structure intrinsic, define the stabilized substrate subspace $\mathcal{H}_s$ and the corresponding orthogonal projector

$$\mathcal{P}_s \equiv \sum_{i \in s} |\phi_i\rangle\langle\phi_i|. \quad (20)$$

For any proto-mode $|\psi_{\text{proto}}\rangle$, the substrate overlap is

$$\epsilon_\psi \equiv \|\mathcal{P}_s|\psi_{\text{proto}}\rangle\|^2 = \langle\psi_{\text{proto}}|\mathcal{P}_s|\psi_{\text{proto}}\rangle. \quad (21)$$

A dimensional illustration of how small overlap coefficients ($\epsilon \ll 1$) naturally generate hierarchically small masses is provided in Supplementary Section 5.

Empirically, long-lived particles occupy the underdamped basin $\chi \ll 1$, while short-lived resonances occupy a secondary $\chi \sim 0.1$ –1 band whose sub-cluster evidence is suggestive rather than conclusive. This distribution is shown in Figure 3; exact values and conventions are in Supplementary Table S1. A formal treatment of the inheritance relation is provided in Ref. [24].

5.2 Particle and substrate classification

For empirical unstable particles, ω is identified with rest mass and Γ with the measured total decay width from PDG 2022 [16]. For stable particles, only upper bounds on Γ are physically meaningful. For confined quarks, any assigned width is a hadronization-scale convention.

Vacuum-scale anchors. Vacuum-scale entries (QCD Scale, Electroweak Scale, GUT Scale, Planck Scale, Dark Energy) are not empirical particle-width measurements. They are defined reference anchors satisfying $\Gamma = 2\omega$ by construction and are excluded from all statistical tests of clustering. The dark energy entry uses the Hubble energy scale as a characteristic frequency proxy; dark energy has no measured decay width, and the assignment $\Gamma = 2\omega_{\text{DE}}$ is imposed by the anchor convention, not measured or derived.

QCD substrate limitation. The assignment $\chi_{\text{QCD}} = 1$ is a phenomenological placeholder. The strong coupling $\alpha_s(Q^2)$ runs from ≈ 0.118 at the Z -pole to non-perturbative values below 1 GeV; any Γ_{QCD} used here denotes an effective scalar-channel scale coarse-grained over non-perturbative dynamics. The appropriate falsification channel is finite-temperature lattice QCD extraction of the scalar spectral peak width $\chi_{\text{lattice}}(T) = \Gamma(T)/(2m_0(T))$ (Supplementary Section 4).

Near-unity empirical candidates and their uncertainties. Among empirical entries, the $\sigma/f_0(500)$ is the closest to $\chi = 1$, with central value $\chi_\sigma = 550/(2 \times 475) \approx 0.58$. Across the full PDG uncertainty range (mass 400–550 MeV, width 400–700 MeV), the stability ratio spans $\chi_\sigma \in [0.36, 0.88]$; $\chi = 1$ is within the uncertainty range but above the central value. This entry is characterized as near-unity within uncertainties, not at unity.

Statistical analysis. A one-sample Kolmogorov-Smirnov test comparing $\log_{10}(\chi)$ for the empirical subset ($n = 23$, vacuum anchors excluded, photon and gluon excluded as massless) against the analytical uniform distribution over the observed range yields $D = 0.540$, $p = 8.39 \times 10^{-7}$ (verified stable across Monte Carlo null samples of 10^5 – 10^7 and the analytical one-sample result). This analysis is not intended to establish a sharply defined cluster at $\chi \approx 1$, but to demonstrate non-uniform structure in χ -space inconsistent with a null distribution. Bootstrap confidence intervals

(10^4 resamples) give median $\log_{10} \chi = -16.9$ (95% CI: $[-33.3, -12.2]$) for stable matter and -1.0 (95% CI: $[-1.9, 1.0]$) for short-lived resonances ($\chi > 0.01$). The non-uniformity reflects two overlapping structures: the dominant concentration of matter at $\chi \ll 1$ (consistent with survivorship of stable structures) and a secondary resonance cluster near $\chi \sim 0.1$ – 1 . Evidence for the resonance sub-cluster alone is characterized as suggestive rather than conclusive.

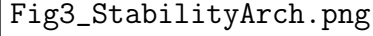


Fig3_StabilityArch.png

Figure 3: **Exceptional-Point Stability Architecture: cross-scale classification.** Physical systems plotted as damping scale $\log_{10} \Gamma$ versus characteristic frequency $\log_{10} \omega$ from cosmological to Planck scales. The solid magenta line marks the $\chi = \Gamma/(2\omega) = 1$ exceptional-point boundary; dashed lines show constant- χ reference levels at $\chi = 0.01, 0.1$, and 10 . For elementary particles, ω is identified with rest mass (natural units) and Γ with the measured decay width [16]. **Open markers** indicate vacuum-substrate anchors defined by the critical-damping condition $\Gamma = 2\omega$ ($\chi = 1$ by construction); these are theoretical reference points, not measured decay widths, and are excluded from all statistical estimates. **Filled markers** represent empirical PDG data. Long-lived matter occupies the deep underdamped basin ($\chi \ll 1$, lower left); short-lived resonances cluster near $\chi \sim 0.1$ – 1 . One-sample KS test versus uniform null: $D = 0.540$, $p = 8.39 \times 10^{-7}$ ($n = 23$, anchors excluded); the non-uniformity reflects survivorship at $\chi \ll 1$ and a secondary resonance cluster; the sub-cluster evidence alone is suggestive rather than conclusive. Exact values and conventions in Supplementary Table S1.

When a dominant substrate satisfies $\chi \approx 1$, any emergent mode with nonzero overlap carries a corresponding near-critical contribution. In the dominance limit, $|c_s|^2 \gg |c_{k \neq s}|^2$, the emergent ratio approaches $\chi_\psi \rightarrow \chi_s$.

5.3 Gauge-coupling interpretation (speculative)

The following interpretation is speculative and is not derived from first principles within the present framework.

Within the stability architecture, gauge couplings are classified as substrate-ratio parameters. In representative open-field settings, interaction-induced broadening rates admit scaling $\Gamma_i \propto g_i^2 \Omega_i$ up to phase-space factors and kinematic thresholds, so the effective stability index $\chi_i = \Gamma_i / (2\Omega_i)$ depends directly on the gauge coupling. The persistence of a gauge sector as a coherent substrate is consistent with operation within the near-critical window $\chi_i \approx 1$. No parameter-free derivation of the numerical values of (g_1, g_2, g_3) is claimed, and this interpretation requires substantial further development before it can be considered predictive.

6 Experimental and Observational Tests

The structural identity of the $\chi = 1$ boundary yields falsifiable predictions across accessible energy scales. Each test specifies an observable, a measurable threshold, and the conditions under which the framework would be challenged.

6.1 Cosmological Synchronization (DESI & Euclid)

The offset Δz (Eq. (13)) is identically zero in flat Λ CDM and generically nonzero in extensions.

- **Observable:** The redshift difference Δz between the kinematic transition ($q = 0$) from supernovae/BAO and the structural transition ($\chi_\delta = 1$) from growth-rate measurements ($f\sigma_8$).
- **Sensitivity:** Upcoming surveys (DESI, Euclid, Rubin) are expected to constrain $z_t \approx 0.6$ with precision $\sigma_z < 0.05$ [8]. A statistically significant $|\Delta z| > 3\sigma_z$ that cannot be attributed to curvature or $w(z)$ evolution would challenge the structural identity.

6.2 Quantum Spectral Merger (Circuit QED)

The circuit QED test is the most near-term experimentally accessible prediction of the framework. Engineered superconducting-circuit platforms can tune effective damping-to-frequency ratios in reduced non-Hermitian response channels [3]. The parameters below describe an illustrative protocol for such an engineered dissipative mode; reaching $\chi \sim 1$ requires a synthetic non-Hermitian two-mode design or direct engineering of the effective decay channel, not a standard weak-Purcell dispersive transmon operating point.

- **Protocol:** A transmon qubit ($\omega_q/2\pi \approx 5$ GHz) coupled to a dissipative cavity ($\kappa/2\pi \approx 1$ MHz) is tuned through the exceptional point by varying the Purcell decay rate $\gamma_P = \kappa g^2/\Delta^2$.
- **Signature:** As χ is scanned through $\{0.8, 0.9, 1.0, 1.1\}$, the qubit population $P_1(t)$ must transition from oscillatory decay $e^{-\gamma t/2} \cos(\omega_a t)$ to the EP2 kernel $te^{-\omega t}$ at the critical point, accompanied by spectral doublet merger $\omega_\pm \rightarrow \omega_0$.
- **Challenge condition:** Failure to observe spectral coalescence within $0.95 < \chi < 1.05$ would challenge the EP2 boundary in open quantum dynamics. Detailed protocols and statistical frameworks are in Supplementary Section 9.

6.3 Finite-Temperature QCD Coherence (Lattice)

- **Observable:** $\chi_{\text{lattice}}(T) = \Gamma(T)/(2m_0(T))$ from finite-temperature scalar 0^{++} channel spectral reconstructions.
- **Signature:** A temperature band near the confinement crossover where $0.8 \lesssim \chi(T) \lesssim 1.2$.
- **Challenge condition:** Robust absence of any near-critical window across all reconstruction methods would weaken the substrate-coherence interpretation.

6.4 Neutron Star Oscillation Modes (Gravitational Waves)

For neutron-star f -modes, $\chi_{\text{NS}} = (4\pi f\tau)^{-1}$. [26] computed f -mode parameters across approximately 9000 EOS configurations, finding $\nu_f \in [1.80, 2.20]$ kHz and $\tau_f \in [0.13, 0.22]$ s for NS masses 1.4 – $2.0 M_\odot$, yielding $\chi_{\text{NS}} \in [1.6, 3.4] \times 10^{-4}$ across the full EOS ensemble. For χ_{NS} to reach unity at $f = 2$ kHz would require $\tau \approx 40 \mu\text{s}$, a factor of 3267 below the shortest observed damping time; no EOS variation closes this gap.

The previously stated prediction that f -mode trajectories approach $\chi_{\text{NS}} \approx 1$ near M_{max} is retracted. The f -mode is classified as occupying the deep underdamped basin, consistent with the broader empirical pattern. A residual falsifiable prediction concerns viscous dissipation channels: equations of state with significant bulk viscosity (hyperon softening, quark-matter deconfinement) shift χ_{NS} upward relative to the gravitational-wave-only prediction. Systematic measurement of $\chi_{\text{NS}}(M)$ with third-generation detectors (Cosmic Explorer, Einstein Telescope) can constrain this contribution; a systematic deviation from the gravitational-wave-only trend would provide evidence for internal dissipation. This is a testable conjecture rather than a prediction of the framework.

7 Conclusion

7.1 Limitations

The following limitations are stated explicitly.

Flat Λ CDM dependence. The identity $\chi_\delta = 1 \iff q = 0$ is exact only for spatially flat geometry, pressureless clustering matter, general relativity, and $w = -1$. Generic extensions produce a nonzero Δz , although specially tuned models could preserve $\Delta z = 0$ accidentally.

Linear response regime. Systems requiring strongly nonlinear, multi-resonance, or non-Markovian descriptions may not admit a meaningful single- χ characterization.

Phenomenological coarse-graining. The particle mapping ($\omega = m$, $\Gamma = \Gamma_{\text{tot}}$) compresses full decay amplitudes, branching ratios, spin structure, and interference effects into two numbers.

Perturbative RG scope. Estimates apply to weakly coupled $\lambda\phi^4$ theory. Non-perturbative regimes and the top-sector Yukawa ($d\chi/d\ln\mu \approx 6 \times 10^{-3}$ per e -fold) are not covered.

Information-efficiency retraction. Full numerical integration of $\eta(\chi)$ for both output-power and kinetic entropy proxies shows no interior maximum near $\chi = 1$ for any tested bandwidth configuration; extrema occur at scan boundaries rather than near the critical-damping point (Supplementary Section 2.4). The original claim $\chi^* \approx 1.02$ is retracted for these proxy definitions. Lemma S2.1 (conditional localization) is retained but does not assert existence of the maximum for the proxies tested.

Substrate inheritance. The inheritance relation is a structural hypothesis motivated by consistency in reduced descriptions; it is not derived from first principles.

Neutron star prediction (retracted). $\chi_{\text{NS}} \in [1.6, 3.4] \times 10^{-4}$ across the [26] EOS ensemble; the original prediction that $\chi_{\text{NS}} \rightarrow 1$ near M_{max} is numerically indefensible by a factor of 3267 and is retracted.

7.2 Summary

The framework is evaluated not by retrospective reinterpretation but by forward falsifiability: each domain discussed yields a concrete test whose failure would directly challenge the proposed structural correspondence.

A single dimensionless ratio, $\chi = \gamma/(2\omega)$, provides a common stability classification language across quantum, field-theoretic, and cosmological dynamics. The critical boundary $\chi = 1$ marks a non-Hermitian EP2 that coincides exactly with the onset of cosmic acceleration via the algebraic identity $\chi_\delta = 1 \iff q = 0$ within flat Λ CDM. Substrate inheritance constrains the admissible parameter space for emergent modes in the leading-order projection approximation, and the observed particle distribution is consistent with stability-based classification. The results range from algebraically exact (Section 3) through structurally motivated (Section 5) to explicitly speculative (Section 5.2). The principal falsifiable tests are: (i) measurement of Δz between the $q = 0$ and $\chi_\delta = 1$ transitions (identically zero in flat Λ CDM and generically nonzero in extensions); (ii) spectral coalescence at $\chi = 1$ in circuit QED platforms, the most near-term accessible test; (iii) near-critical scalar-channel width in finite-temperature lattice QCD near the confinement crossover; and (iv) systematic measurement of $\chi_{\text{NS}}(M)$ in gravitational-wave observations, with the viscous-dissipation channel as the falsifiable target. The framework is therefore offered as a structurally grounded classification with defined scope and testable consequences.

Methods

Settling-time simulation. Numerical settling times were computed by integrating the second-order damped oscillator

$$\ddot{x} + \gamma\dot{x} + \omega^2x = 0 \tag{22}$$

via `scipy.integrate.solve_ivp` (Python 3.9+, RK45 adaptive time stepping, absolute and relative tolerances 10^{-9} , maximum step $\Delta t = 0.01$) with unit frequency $\omega = 1$, initial conditions $(x_0, \dot{x}_0) = (1, 0)$, and a damping grid of 120 uniformly spaced values $\chi \in [0.10, 3.0]$, corresponding to $\gamma \in [0.20, 6.0]$. Settling time T_s was defined as the first time at which $|x(t')| \leq 10^{-3}$ for all $t' \geq t$. Results are archived at Zenodo (<https://doi.org/10.5281/zenodo.17528111>).

Statistical null tests. A one-sample Kolmogorov-Smirnov test was applied to the empirical $\log_{10}(\chi)$ distribution ($n = 23$, vacuum anchors excluded) against the analytical uniform distribution over the observed range, yielding $D = 0.540$, $p = 8.39 \times 10^{-7}$ (verified stable across Monte Carlo

null samples 10^5 – 10^7 and the analytical result). Bootstrap confidence intervals used 10^4 resamples. The transition redshift $z_t = 0.632 \pm 0.018$ (1σ) was computed by propagating the Planck 2018 uncertainty $\sigma_{\Omega_m} = 0.007$ via Monte Carlo over 2×10^5 samples. Source code archived at Zenodo (<https://doi.org/10.5281/zenodo.17528111>).

Cosmological calculations. The identity $\chi_\delta = 1 \iff q = 0$ was derived analytically from the flat- Λ CDM Friedmann and growth equations (Section 3 and Supplementary Section 1.3). Figure 2(a) plots $q(z)$ and $\chi_\delta(z)$ computed directly from these equations; no fitting functions are used. Figure 2(b) plots the trajectory in (χ_δ, q) space for Λ CDM and representative w CDM models. Concordance parameters: $\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$, $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [19]. Sensitivity of the synchronization redshift to w CDM departures was computed over a grid of (w_0, w_a) values; results tabulated in `phase1_deltaa_grid.csv` and `phase1_validity_table.csv`, provided in the Supplementary Materials.

Particle data compilation. Mass and width values in Supplementary Table S1 were drawn from PDG 2022 [16]. Vacuum-scale entries are defined anchors ($\chi = 1$ by construction, $\Gamma = 2\omega$), plotted as open markers in Figure 3 and excluded from all statistical estimates. Quark widths represent the hadronization scale $\sim \Lambda_{\text{QCD}}$. Electron and proton widths are PDG upper bounds. Neutrino entries report mass scales only; coherence constraints imply $\chi_k \ll 1$. Figure 3 generated from `stability_table_S1.csv` using `figure3_clean.py`; both archived at Zenodo (<https://doi.org/10.5281/zenodo.17528111>).

Author Contributions

N.C. conceived the theoretical framework, performed the derivations, conducted the analysis, and prepared the manuscript and supplementary materials.

Acknowledgments

The author acknowledges the broader scientific community for prior advances that made this work possible.

Funding

This research received no external funding.

Competing Interests

The author declares no competing interests.

Data Availability

The settling-time simulation data (`symc_settling_times.csv`), cosmological sensitivity grids (`phase1_deltaa.csv`, `phase1_validity_table.csv`), canonical particle data file (`stability_table_S1.csv`), and complete source code (`figure3_clean.py`, `chi_null_tests.py`) are available at Zenodo (<https://doi.org/10.5281/zenodo.17528111>) and GitHub (<https://github.com/SymCUniverse/Foundations>). All particle physics values are drawn from PDG 2022 [16].

AI Writing Assistance

During the preparation of this work, the author used Claude (Anthropic), GPT-4 (OpenAI), DeepSeek, Gemini (Google), and Microsoft Copilot in a systematic convergence methodology to cross-check mathematical derivations, test theoretical claims, and refine hypothesis formulation through iterative dialogue. Multiple AI systems were queried independently on identical problems to identify consensus solutions and flag inconsistencies requiring further investigation. The author orchestrated all queries, evaluated all outputs critically, and edited all content for accuracy and clarity. All data analysis, scientific conclusions, theoretical framework development, and final manuscript content are the original intellectual contribution of the author. The author takes full responsibility for the accuracy and integrity of this work.

References

- [1] Bender, C. M. Making sense of non-Hermitian Hamiltonians. *Rep. Prog. Phys.* **70**, 947–1018 (2007).
- [2] Aspelmeyer, M., Kippenberg, T. J. & Marquardt, F. Cavity optomechanics. *Rev. Mod. Phys.* **86**, 1391–1452 (2014).
- [3] Blais, A. et al. Circuit quantum electrodynamics. *Rev. Mod. Phys.* **93**, 025005 (2021).
- [4] Breuer, H.-P. & Petruccione, F. *The Theory of Open Quantum Systems* (Oxford University Press, 2002).
- [5] Cardoso, V., Franzin, E. & Pani, P. Is the gravitational-wave ringdown a probe of the event horizon? *Phys. Rev. Lett.* **116**, 171101 (2016).
- [6] Carmichael, H. J. *Statistical Methods in Quantum Optics 1* (Springer, 1999).
- [7] Clerk, A. A. et al. Introduction to quantum noise, measurement, and amplification. *Rev. Mod. Phys.* **82**, 1155–1208 (2010).
- [8] DESI Collaboration. DESI 2024 VI: cosmological constraints from BAO. Preprint at <https://arxiv.org/abs/2404.03002> (2024).
- [9] Dodelson, S. *Modern Cosmology* (Academic Press, 2003).
- [10] Gorini, V., Kossakowski, A. & Sudarshan, E. C. G. Completely positive dynamical semigroups of N-level systems. *J. Math. Phys.* **17**, 821–825 (1976).
- [11] Heiss, W. D. The physics of exceptional points. *J. Phys. A* **45**, 444016 (2012).
- [12] Kato, T. *Perturbation Theory for Linear Operators* (Springer, 1995).
- [13] Bergholtz, E. J., Budich, J. C. & Kunst, F. K. Exceptional topology of non-Hermitian systems. *Rev. Mod. Phys.* **93**, 015005 (2021).
- [14] Lindblad, G. On the generators of quantum dynamical semigroups. *Commun. Math. Phys.* **48**, 119–130 (1976).
- [15] Ogata, K. *Modern Control Engineering* 5th edn (Prentice Hall, 2010).

- [16] Particle Data Group. Review of particle physics. *Prog. Theor. Exp. Phys.* **2022**, 083C01 (2022).
- [17] Peebles, P. J. E. *Principles of Physical Cosmology* (Princeton University Press, 1993).
- [18] Perlmutter, S. et al. Measurements of Ω and Λ from 42 high-redshift supernovae. *Astrophys. J.* **517**, 565–586 (1999).
- [19] Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. *Astron. Astrophys.* **641**, A6 (2020).
- [20] Riess, A. G. et al. Observational evidence from supernovae for an accelerating universe. *Astron. J.* **116**, 1009–1038 (1998).
- [21] Abbott, B. P. et al. GW190814: gravitational waves from the coalescence of a 23 solar mass black hole with a 2.6 solar mass compact object. *Astrophys. J. Lett.* **896**, L44 (2020).
- [22] Wilson, K. G. & Kogut, J. The renormalization group and the ϵ expansion. *Phys. Rep.* **12**, 75–199 (1974).
- [23] Christensen, N. Structural constraints from critical damping in open quantum field theories: implications for QCD substrate inheritance and phenomenological extensions. Zenodo <https://doi.org/10.5281/zenodo.17437688> (2026).
- [24] Christensen, N. Closing critical gaps: physical inheritance from stabilized substrates in dynamical systems. Zenodo <https://doi.org/10.5281/zenodo.17428940> (2026).
- [25] Wang, F.-Y. & Dai, Z.-G. Constraining dark energy and cosmological transition redshift with Type Ia supernovae. *Chin. J. Astron. Astrophys.* (2006). Preprint at <https://arxiv.org/abs/0708.4062>.
- [26] Guha Roy, D., Malik, T., Bhattacharya, S. & Banik, S. Analysis of neutron star f -mode oscillations in general relativity with spectral representation of nuclear equations of state. *Astrophys. J.* **968**, 124 (2024). Preprint at <https://arxiv.org/abs/2312.02061>.